Monsoon, rainfall, weather, and climate

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Introduction

➢ Climate is the long-term average of weather conditions in a region, usually measured over 30+ years.

➢ Key Elements: It includes temperature, humidity, precipitation, wind, and atmospheric pressure.

➢ Difference from Weather: Unlike daily-changing weather, climate describes consistent patterns that influence nature and human life.

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Director,

Department of Employment and Training.

Factors Influencing Climate

Latitude

India's location from 8°4'N to 37°6'N leads to climatic diversity.
South of Tropic of Cancer: Tropical, hot year-round.
North of Tropic: Subtropical, with clear seasonal variation.

Altitude

Temperature decreases by 6.5°C per 1000 m altitude.
Mountainous regions are cooler than plains.

Distance from Sea

Coastal regions: Moderate temperatures, high humidity and rainfall.
Inland regions: Extreme temperature fluctuations, less rainfall.

Monsoon Winds

Seasonal reversal: Southwest monsoon (heavy rain) and northeast monsoon (affects SE India).
Responsible for India's equable (moderate) climate.

Relief (Topography)

Himalayas block cold winds → warm northern winters.
Windward vs. leeward effect: Mangalore (280 cm rain), Bengaluru (50 cm).

Jet Streams

High-altitude winds that influence the onset of the monsoon.
Subtropical westerly jet is critical in shifting monsoon systems.

Monsoon System

"Monsoon" = season (Arabic origin).
Brings summer rain from SW, reverses in winter.
Central to India's seasonal climate pattern.

Climate of India

Climate Type & Seasons

- India has both tropical and subtropical climates.
- Four major seasons:
  - Winter (Jan–Feb)
  - Summer (Mar–May)
  - Southwest Monsoon (Jun–Sep)
  - Post-monsoon / Northeast Monsoon (Oct–Dec)

Rainfall Patterns

- Monsoons create distinct wet and dry seasons.
- Rainfall varies extremely:
  - **Mawsynram & Cherrapunji:** ~1100 cm (wettest)
  - **Jaisalmer & Ladakh:** ~12 cm (driest)
- Southwest Monsoon (June–Sept) brings rain to most regions like the Ganga delta and Odisha.
- The Coromandel Coast remains mostly dry during this and gets rain from the Northeast Monsoon (Oct–Dec).
- Monsoon reaches Goa/Hyderabad/Patna by early June, and northwest India by early July.

Temperature Variations

- Thar Desert has scorching days and cold nights.
- Himalayan regions face extreme cold in winter, mild summers.
- Coastal areas have minimal temperature variation due to the ocean.
- In December, Jammu & Kashmir can drop to -40°C, while coastal areas remain at 20–25°C.
- Summers are intensely hot in plains and deserts; winters are moderate to severely cold depending on the region.

Seasons of India

Meteorologists recognize four distinct seasons in India:

- **Winter Season** (January – February)
- **Summer Season** (March – May)
- **Southwest Monsoon or Rainy Season** (June – September)
- **Northeast Monsoon Season** (October – December)

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Winter Season (January – February)

- Sun's rays fall on the Tropic of Capricorn, resulting in slanting rays over India and cool temperatures.
- Clear skies, low humidity, and large day-night temperature variations are common.
- Rainfall occurs in:
  - Western Himalayas, Tamil Nadu, Kerala, aiding winter crop cultivation.
- Pressure Pattern:
  - High pressure in the north; low pressure in the south.
  - Winds flow from NW to SE, but with low velocity.
- Western Disturbances (from the Mediterranean) bring:
  - Rain to Indus-Ganga plains, snowfall in Himalayas, cold waves and fog.
- Tropical Cyclones:
  - Rare, but some form in the Bay of Bengal and impact Tamil Nadu.
- Winter Monsoon Rain:
  - Affects Tamil Nadu, Andhra Pradesh, Karnataka, and Kerala.

Summer Season (March – May)

- Sun's rays shift northward, increasing temperatures from south to north.
- Very high temperatures in central and north India:
  - Southern India: 40–45°C
  - Rajasthan: Up to 48°C
  - Record: Alwar (50.5°C), Ganganagar (50.6°C)
- Coastal areas remain cooler due to maritime winds.
- Heatwaves:
  - Frequent in Rajasthan, Punjab, UP, Bihar.
  - Rare below 13°N latitude due to sea influence.
- Low pressure develops, especially in NW India, delaying monsoon.
- Hot winds:
  - Loo: Dry, dusty winds (30–40 km/h) blow across NW India.
  - Andhis: Dust storms in north India bring temporary cooling.
- Thunderstorms:
  - Norwesters (Kalbaisakhi) in Bengal & Assam — help tea/jute crops.
  - Mango showers in Kerala/Karnataka aid fruit ripening.
- Cyclones:
  - Frequent in Bay of Bengal, rare in Arabian Sea.
- Rainfall is limited, mostly in NE India and Kashmir (via western disturbances).

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Southwest Monsoon (June–September)

Key Features

- Onset: Begins in Kerala in early June; covers entire India by mid-July.
- Rainfall: Contributes 75% of India's annual rainfall.
- Burst: Arrives with thunder and lightning (called "burst of monsoon").

Two Branches

- Arabian Sea Branch:
  - Hits Western Ghats → heavy orographic rainfall (400–500 cm).
  - Leeward areas get little rain (30–50 cm); Rajasthan receives less due to Aravalli alignment.
  - Streams: Western Ghats, Central India (Narmada/Tapti), Rajasthan (low rainfall except Mt. Abu).
- Bay of Bengal Branch:
  - NE India stream: Heavy rain in Mawsynram & Cherrapunji.
  - Ganga Plain stream: Rain decreases from east to west.
  - Tamil Nadu stays dry due to rain shadow effect and parallel winds.

Breaks in Monsoon

- Occur in July–August; cause temporary dry spells.
- Caused by:
  - Northward shift of monsoon trough (ITCZ).
  - Cyclones drawing away moisture.

El Niño Effect

- El Niño warms Pacific Ocean → weakens Indian monsoon → drought.
- In winter El Niño, more cyclones form in Bay of Bengal.

Northeast Monsoon (October–December)

Formation

- Occurs after SW monsoon withdrawal.
- Wind reversal due to southward shift of pressure belts.
- Winds blow from northeast over Bay of Bengal → pick up moisture.

Rainfall Distribution

- Brings rain to Tamil Nadu (35% of annual), Andhra Pradesh, Kerala, and South Interior Karnataka.
- Linked with Bay of Bengal cyclones and depressions.

Weather Conditions

- Intense rainfall, strong winds, and coastal flooding.
- Drop in temperatures marks start of cooler weather.

Importance

- Crucial for Tamil Nadu agriculture, especially where SW monsoon is weak.
- Cyclones during this period can cause floods and severe damage.

Monsoon Winds and Seasons

Monsoons dominate India's climate, causing seasonal wind reversal and rainfall.

Two main types:

- **Southwest Monsoon (June–Sept):** Brings heavy rain across most of India.
- **Northeast Monsoon (Oct–Dec):** Brings rain mainly to **Tamil Nadu** and **SE Andhra Pradesh.**

Monsoon Characteristics

- Onset: Starts in Kerala (early June) with thunder, lightning, and heavy rain.
- Spread: Covers India gradually.
- Retreat: Begins from North India in late September, moving southward.

Jet Streams' Role

- Westerly Jet (Winter): Brings Western Disturbances, causing rain/snow in north India.
- Easterly Jet (Summer): Forms due to Tibetan Plateau heating, triggers SW monsoon onset.

Tropical Cyclones and Western Disturbances

Tropical Cyclones	Form in **Bay of Bengal** and **Arabian Sea.**
	Strongest during **retreating monsoon (Oct–Nov);** mainly affect **east coast.**
Western Disturbances	Come from the **Mediterranean Sea.**
	Cause **winter rain and snow** in **North India and Himalayas.**

El Niño, La Niña, and ENSO

El Niño

* Warming of central/eastern Pacific Ocean.
* Weakens Indian monsoon, causes low rainfall and drought.
* Reduces cyclone formation in Bay of Bengal.

La Niña

* Cooling of Pacific waters (opposite of El Niño).
* Strengthens monsoon, causes heavy rain and flood risk.
* More cyclones in Bay of Bengal.

ENSO

* Interaction between ocean temperatures and pressure systems in the Pacific.
* During El Niño + Southern Oscillation, monsoon weakens, causing dry conditions in India.

The Indian Monsoon: Formation, Influencing Factors, and Theories

Southwest Monsoon – Formation & Factors

- Tibetan Plateau Heating: Creates low pressure, pulls moist winds.
- High Pressure near Madagascar: Pushes moisture-laden winds toward India.

Onset Influencers:

- Subtropical Jet Stream (STJ) shifts north.
- Tropical Easterly Jet brings ocean moisture.
- ITCZ moves north, guiding monsoon winds into India.

Intensity Influencers:

- Strength of pressure systems.
- Somali Jet & Current carry moisture.
- Walker Circulation & Indian Ocean Dipole affect rainfall spread.

Northeast Monsoon – Formation

- High-pressure zones over Tibet and Siberia push dry winds south.
- ITCZ shifts south, helping form northeast monsoon.
- Affects Tamil Nadu and southern India during Oct–Dec.

Theories Explaining Indian Monsoons

Classical Theory – Sir Edmund Halley

- Monsoon is like a giant land-sea breeze
- Summer: Land heats faster → low pressure → moist air from sea → rain.
- Winter: Land cools faster → high pressure → air flows to sea → dry winds (NE monsoon)
- Limitation: Doesn't explain monsoon variability, bursts, or delays.

Modern Theories

- Include jet streams, ocean currents, and global climatic phenomena
- El Niño & La Niña
- Indian Ocean Dipole (IOD)
- Walker Circulation

Distribution of Rainfall and Heatwaves in India

Rainfall Distribution in India

- Average Rainfall: ~118 cm annually; uneven across regions.

Rainfall Zones:

- Heavy Rainfall (>200 cm): Western coast, Assam, Meghalaya, Tripura, Nagaland, Arunachal.
- Low Rainfall (<100 cm): Rajasthan, Punjab, Haryana, Deccan Plateau (except TN coast).
- Moderate Rainfall (100–200 cm): Rest of India.

Key Influencing Factors:

- Moisture-laden Westerlies from Arabian Sea.
- Monsoon Trough shift:
  - Southward = More rain in central/south India.
  - Northward = More rain in Himalayas.
- Offshore Trough & Wind Shear Zone boost coastal and central rainfall.
- Cyclonic Systems from Bay of Bengal bring rain inland.

Heatwaves in India

Definition & Criteria (IMD):

- High temperatures, mostly during March–June.
- Heatwave if:
  - Plains ≥ 40°C, Hilly ≥ 30°C, Coastal ≥ 37°C.
  - Deviation of 5–7°C above normal or >45°C overall.

Causes:

- Hot, dry air & clear skies.
- Lack of moisture.
- Anticyclones trap heat.
- Climate change & vulnerable geography.

Impacts:

- Health issues: 730 deaths (2023).
- Water shortage, power demand surge.
- Crop damage, wildfires.

Climate of Tamil Nadu

General Climate	• Tropical climate with high temperatures year-round. • Experiences three rainfall periods and a distinct dry season.
Rainfall Patterns	• Advanced Rainfall – Pre-monsoon showers before the main monsoon. • Tropical Cyclone Rainfall – Oct–Nov cyclones from the Andaman Sea bring rain. • Northeast Monsoon (Oct–Dec) – Tamil Nadu's main rainy season, due to northeast winds and Bay of Bengal disturbances.
Dry Season	• Feb to early June: Hot and dry. • Monsoon period: Mid-June to December, refills water sources. • Monsoon failure causes droughts due to rain dependence.
Agro-Climatic Zones	• North-East Zone • North-West Zone • Western Zone

	• Southern Zone • High Rainfall Zone • High Altitude Hilly Zone • Cauvery Delta Zone (most fertile agricultural zone)
Geographical Influence	• Near Equator → Receives vertical sun rays, stays hot. • East Coast → Tropical maritime climate (moderated by sea). • Western Hills → Cooler due to elevation. • Interior Areas → Hotter and drier due to low altitude and sea distance.

Seasons of Tamil Nadu

Tamil Nadu has **4 distinct seasons**, influenced by the **Sun's position**:

Season	**Period**
Winter	January – February
Summer	March – May
Southwest Monsoon	June – September
Northeast Monsoon	October – December

Winter (Jan–Feb)

- Sun's rays are slanting, resulting in cooler temperatures.
- Plains: 15°C–25°C; Hills: Below 5°C (Nilgiris may hit 0°C).
- Weather: Cold mornings, mist, dry conditions, no major rainfall.

Summer (Mar–May)

- Sun moves north, Tamil Nadu receives direct vertical rays.
- Temps range from 30°C to 40°C, sometimes higher.
- Weather: Hot, dry, occasional dust storms.
- Pre-monsoon showers (Mango Showers) in May.
- Convectional rain occurs in some areas.

Monsoons in Tamil Nadu

Tamil Nadu receives rainfall from **two monsoons**.

Southwest Monsoon (June–September)	Northeast Monsoon (Oct–Mid-December)
➢ Formed due to low pressure over North India in summer. ➢ Brings heavy rain to most of India, but minimal rain to Tamil Nadu. Why Less Rain in TN? ➢ Western Ghats rain shadow effect blocks Arabian Sea winds. ➢ Rainfall Distribution: o Coimbatore Plateau: ~50 cm o Kanyakumari, Nilgiris, Tirunelveli: 50–100 cm o Eastern Tamil Nadu: Very scanty rain	➢ Caused by cold, dry winds from Central Asia, deflected toward Bay of Bengal. ➢ Known as "Retreating Monsoon". ➢ Main rainy season for Tamil Nadu, contributing 48% of annual rainfall. Rainfall Distribution: ➢ Coastal districts: ~60% of annual rain ➢ Interior TN: 40–50% ➢ East coastal TN: 100–200 cm ➢ Central/Northwest TN: 50–100 cm

Role of Cyclones

- Frequent Bay of Bengal cyclones bring heavy rain to coastal areas.
- Cyclones contribute to 50%+ of TN's rainfall, but also cause floods and crop damage